

Math 526, S05
Exam 2

Name:
SSN: xxx-xx-

Instructions:

- There are totally 6 problems.
- You must show all of your work to receive a credit for a correct answer.
- No graphical calculator is allowed in the quiz and exam.

Problem	Points	Score
1	15	
2	25	
3	15	
4	10	
5	20	
6	15	
Total	100	

Good Luck !

Problem 1. (15 points) Let $\mathbf{x} = \begin{bmatrix} 2 \\ -5 \\ 4 \\ 2 \end{bmatrix}$, $\mathbf{A} = \begin{bmatrix} 2 & 4 & -4 \\ -1 & -2 & 5 \\ 2 & 4 & -4 \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} -2 & 3 \\ 1 & -1 \end{bmatrix}$.

a) Compute $\|\mathbf{x}\|_1$, $\|\mathbf{x}\|_2$, and $\|\mathbf{x}\|_\infty$.

Solution:

$$\|\mathbf{x}\|_1 = 2 + 5 + 4 + 2 = 13$$

$$\|\mathbf{x}\|_2 = \sqrt{2^2 + (-5)^2 + 4^2 + 2^2} = 7$$

$$\|\mathbf{x}\|_\infty = \max\{2, 5, 4, 2\} = 5$$

b) Compute $\|\mathbf{A}\|_1$ and $\|\mathbf{A}\|_\infty$.

Solution:

$$\|\mathbf{A}\|_1 = \max\{2 + 1 + 2, 4 + 2 + 4, 4 + 5 + 4\} = 13$$

$$\|\mathbf{A}\|_\infty = \max\{2 + 4 + 4, 1 + 2 + 5, 2 + 4 + 4\} = 10$$

b) Compute $\kappa_\infty(\mathbf{B})$.

Solution:

$$\mathbf{B}^{-1} = \begin{bmatrix} 1 & 3 \\ 1 & 2 \end{bmatrix}$$

So

$$\kappa_\infty(\mathbf{B}) = \|\mathbf{B}\|_\infty \cdot \|\mathbf{B}^{-1}\|_\infty = 4 \cdot 5 = 20$$

Problem 2. (25 points) Consider the system $\mathbf{Ax} = \mathbf{b}$ where

$$\mathbf{A} = \begin{bmatrix} 5 & -2 & 1 \\ 1 & 4 & 1 \\ -2 & 1 & -5 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 5 \\ 3 \\ 12 \end{bmatrix}$$

a) Write down the corresponding iterative formula for the Jacobi method, then compute the approximate solutions $\mathbf{x}_1$, $\mathbf{x}_2$ and $\mathbf{x}_3$ of the system by the Jacobi method with the initial guess $\mathbf{x}_0 = \mathbf{0}$.

Solution: The iterative formula for the Jacobi method is

$$\begin{aligned} x_1^{new} &= \frac{1}{5}(5 + 2x_2^{old} - x_3^{old}) \\ x_2^{new} &= \frac{1}{4}(3 - x_1^{old} - x_3^{old}) \\ x_3^{new} &= -\frac{1}{5}(12 + 2x_1^{old} - x_2^{old}) \end{aligned}$$

With $\mathbf{x}_0 = [0 \ 0 \ 0]^t$, we get

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 3/4 \\ -12/5 \end{bmatrix}$$

$$\mathbf{x}_2 = \begin{bmatrix} 1.78 \\ 1.1 \\ -2.65 \end{bmatrix}$$

$$\mathbf{x}_3 \approx \begin{bmatrix} 1.97 \\ 0.9675 \\ 0.2892 \end{bmatrix}$$

b) Write down the corresponding iterative formula for the Gauss-Seidel method, then compute the approximate solutions $\mathbf{x}_1$, $\mathbf{x}_2$ and $\mathbf{x}_3$ of the system by the Gauss-Seidel method with the initial guess $\mathbf{x}_0 = \mathbf{0}$.

Solution: The iterative formula for the Gauss-Seidel method is

$$\begin{aligned}x_1^{new} &= \frac{1}{5}(5 + 2x_2^{old} - x_3^{old}) \\x_2^{new} &= \frac{1}{4}(3 - x_1^{new} - x_3^{old}) \\x_3^{new} &= -\frac{1}{5}(12 + 2x_1^{new} - x_2^{new})\end{aligned}$$

With $\mathbf{x}_0 = [0 \ 0 \ 0]^t$, we get

$$\begin{aligned}\mathbf{x}_1 &= \begin{bmatrix} 1 \\ 1/2 \\ -27/10 \end{bmatrix} \\ \mathbf{x}_2 &\approx \begin{bmatrix} 1.74 \\ 0.99 \\ -2.89 \end{bmatrix} \\ \mathbf{x}_3 &\approx \begin{bmatrix} 1.9756 \\ 0.9806 \\ -2.9941 \end{bmatrix}\end{aligned}$$

c) Which one of these two methods generated a better approximate solution $\mathbf{x}_3$ (in the $\|\cdot\|_\infty$ norm) if we compare them with the exact solution $\mathbf{x} = [2 \ 1 \ -3]^t$?

Solution:

For the Jacobi method:

$$\|\mathbf{x} - \mathbf{x}_3\|_\infty \approx 0.108$$

For the Gauss-Seidel method:

$$\|\mathbf{x} - \mathbf{x}_3\|_\infty \approx 0.0244$$

So the Gauss-Seidel method generated a better approximate solution

Problem 3. (15 points) How many steps are needed to reduce the initial solution error (in the $\|\cdot\|_\infty$ norm) by a factor of 10^{-6} if the Jacobi's method is applied to the following system?

$$\begin{bmatrix} 5 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 5 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 5 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 5 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 5 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 5 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 5 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \\ x_8 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

Solution: By the Jacobi method, we have

$$\mathbf{S}^{-1}\mathbf{T} = \begin{bmatrix} 0 & -1/5 & 0 & 0 & -1/5 & 0 & 0 & 0 \\ 0 & 0 & -1/5 & 0 & 0 & -1/5 & 0 & 0 \\ 0 & 0 & 0 & -1/5 & 0 & 0 & -1/5 & 0 \\ 0 & 0 & 0 & 0 & -1/5 & 0 & 0 & -1/5 \\ -1/5 & 0 & 0 & -1/5 & 0 & 0 & 0 & 0 \\ 0 & -1/5 & 0 & 0 & -1/5 & 0 & 0 & 0 \\ 0 & 0 & -1/5 & 0 & 0 & -1/5 & 0 & 0 \\ 0 & 0 & 0 & -1/5 & 0 & 0 & -1/5 & 0 \end{bmatrix}$$

So

$$\|\mathbf{S}^{-1}\mathbf{T}\|_\infty = 0.4$$

Then we need to solve k for

$$\frac{\|\mathbf{e}_k\|_\infty}{\|\mathbf{e}_0\|_\infty} \leq 10^{-6}$$

where $\mathbf{e}_k = \mathbf{x}_k - \mathbf{x}_0$. But we know

$$\frac{\|\mathbf{e}_k\|_\infty}{\|\mathbf{e}_0\|_\infty} \leq \|\mathbf{S}^{-1}\mathbf{T}\|_\infty^k$$

So we only need to solve

$$\frac{\|\mathbf{e}_k\|_\infty}{\|\mathbf{e}_0\|_\infty} \leq (0.4)^k \leq 10^{-6}$$

which gives

$$k \geq \frac{\lg(0.4)}{\lg(10^{-6})} \geq 16.$$

Problem 4. (10 points) Let $\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 1 & -1 & 4 & -9 \\ 0 & 1 & 0 & 0 \\ 3 & 2 & 7 & -4 \end{bmatrix}$. Compute $\det(\mathbf{A})$. Is $\mathbf{A}$ singular?

Solution: Since the 4th row of $\mathbf{A}$ is the sum of 1st, 2nd and 3rd rows, by the properties of determinants, we have

$$\det(\mathbf{A}) = 0.$$

Consequently, we know that $\mathbf{A}$ is singular.

Problem 5. (20 points)

a) Does the Gauss-Seidel method converge for the system $\begin{bmatrix} 1 & -0.4 \\ 2 & 1 \end{bmatrix} \mathbf{x} = \mathbf{b}$? Explain.

Solution: Yes. For the Gauss-Seidel method, we have

$$\mathbf{S} = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \quad \mathbf{T} = \begin{bmatrix} 0 & 0.4 \\ 0 & 0 \end{bmatrix}$$

and

$$\mathbf{S}^{-1}\mathbf{T} = \begin{bmatrix} 0 & 0.4 \\ 0 & -0.8 \end{bmatrix}$$

So we have

$$\|\mathbf{S}^{-1}\mathbf{T}\|_{\infty} = 0.8 < 1$$

which means that the Gauss-Seidel method converges for $\mathbf{A}$.

b) Is it correct that $\det \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{bmatrix} = \det \begin{bmatrix} d & b & a & c \\ -h & -f & -e & -g \\ -l & -j & -i & -k \\ p & n & m & o \end{bmatrix}$? Explain. (Use the properties of determinants)

Solution: Yes.

$$\begin{aligned} \det \begin{bmatrix} d & b & a & c \\ -h & -f & -e & -g \\ -l & -j & -i & -k \\ p & n & m & o \end{bmatrix} &= -\det \begin{bmatrix} d & b & a & c \\ h & f & e & g \\ -l & -j & -i & -k \\ p & n & m & o \end{bmatrix} = \det \begin{bmatrix} d & b & a & c \\ h & f & e & g \\ l & j & i & k \\ p & n & m & o \end{bmatrix} \\ &= -\det \begin{bmatrix} a & b & d & c \\ e & f & h & g \\ i & j & l & k \\ m & n & p & o \end{bmatrix} = \det \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{bmatrix} \end{aligned}$$

Problem 6. (15 points) True or False (no explanation needed).

a) Let $\mathbf{x}$ be a column 4-vector, then $\|\mathbf{x}\|_1 \leq 2\|\mathbf{x}\|_2$.

True

b) Let $\mathbf{A}$ be a 3×3 matrix, then $\kappa_1(\mathbf{A}) \leq 3\kappa_\infty(\mathbf{A})$.

False

c) Let $\mathbf{A} = \begin{bmatrix} -8 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 3 \end{bmatrix}$, then $\kappa_2(\mathbf{A}) = 3$.

False

d) There is a 5×5 matrix $\mathbf{A}$ such that $\kappa_\infty(\mathbf{A}) = 0.8$.

False

f) $\mathbf{A}$ and $\mathbf{B}$ are two $n \times n$ matrix, we have $\det(\mathbf{AB}) = \det(\mathbf{A}^t\mathbf{B}^t)$.

True